

Intern Technical Assessment

Project Overview

This assessment evaluates your ability to integrate clinical data from EMRs with clinical intelligence engines. You will be building a prototype that extracts patient data and applies clinical logic for ASAM Level of Care (LoC) estimation and The Joint Commission (TJC) compliance auditing.

If you don't understand any of what this means, that's perfectly fine. We felt the same a few months ago. Just do some research before you write code to understand this vertical and some terminologies. That's also a big part of this assessment. Feel free to reach out if you don't understand certain concepts or requirements.

The requirements are vague on purpose. You must find and research data sources if not specified.

Tech Stack Requirements

- The solution should be built using FastAPI, and while a PostgreSQL database is preferred, the database choice is flexible.
- A UI would be nice, but not required.

Task 1: Environment Setup & Data Generation

Create a sandbox environment in [SimplePractice](#) to simulate a realistic clinical workflow. You must populate the chart with the following "synthetic" data for a single patient:

- **Admission Assessment:** A comprehensive Biopsychosocial (BPS) intake that includes history of present illness, substance use history, and initial risk screening.

- **Progress Notes:** Generate at least three progress notes on different dates using different structured formats:
 - **SOAP:** Subjective, Objective, Assessment, Plan.
 - **DAP:** Data, Assessment, Plan.
 - **DSAP:** Data, Subjective, Assessment, Plan.

Task 2: Data Extraction API

Develop a Python-based API that programmatically extracts information from your SimplePractice sandbox. The output should be a structured JSON object containing:

- Patient demographic data.
- The full text of the initial assessment.
- Timeline-ordered progress notes with their respective metadata (date, author, type).

Task 3: Clinical Intelligence Endpoints

Create an API endpoint that processes the extracted patient data to answer specific clinical and regulatory questions.

Endpoint 1: ASAM Level of Care Prediction

Question: According to the present documents, what ASAM level of care should the patient be in?

Your model must evaluate the patient across the six ASAM Dimensions, and you must use ASAM 4th edition for this.

Endpoint 2: Joint Commission (TJC) Compliance Audit

Question: According to the present documents, is the provider meeting all of the [Joint Commission Standards \(CTS\)](#) for behavioral health treatment?

Evaluation Criteria

1. **Extraction Accuracy:** Does the API capture all relevant clinical fields from the SimplePractice forms?
2. **Rationale Quality:** The system should not just provide a "Yes/No" but a brief rationale referencing specific text in the notes (e.g., "Standard CTS.02.02.01 EP2 failed because spiritual orientation was not documented").
3. **ASAM Consistency:** Is the predicted Level of Care (e.g., 2.1 Intensive Outpatient) logically supported by the risk ratings in the dimensions?

Submission Instructions

- Submit your code repository link and a sample JSON response for your simulated patient to both kyle@perspectiveshealth.ai and eshan@perspectiveshealth.ai.
- We will know if you have put in enough effort or not based on the way you think and approach the problem. Don't try to just create a simple wrapper; you will quickly learn you can't do that.